

$$\gcd(a, b) = d \Rightarrow a = d h_1, b = d h_2 \\ (h_1, h_2) = 1$$

$$a - b = d(h_1 - h_2) \Rightarrow \gcd(a, a - b) = \\ = \gcd(\underline{d h_2}, \underline{d(h_1 - h_2)}) = d$$

$$\gcd(18, 24) = \gcd(24, 18) = \gcd(18, 24 - 18) = \\ = \gcd(18, 6) = \gcd(6, 18 - 6) = \gcd(6, 12) = \\ = \gcd(12, 6) = \gcd(6, 12 - 6) = \gcd(6, 6) = 6$$

17	3
14	3
11	3
8	3
5	3
2	3

→

$$\begin{array}{r} 17 \% 3 \quad 3 \\ \hline 2 \quad 3 \end{array}$$

~~$$18 = 0 \cdot 24 + 18$$~~

$$24 = 1 \cdot 18 + 6$$

$$18 = 6 \cdot 3 + 0$$

a' b r

$$18 = 24 \cdot 0 + 18$$

$$24 = 18 \cdot 1 + 6$$

$$32 = 5 \cdot 6 + 2$$

$$5 = 2 \cdot 2 + 1$$

$$2 = 1 \cdot 2 + 0$$

c mod c

while b != 0 {

$$r = a \% b$$

$$a = b$$

$$b = r$$

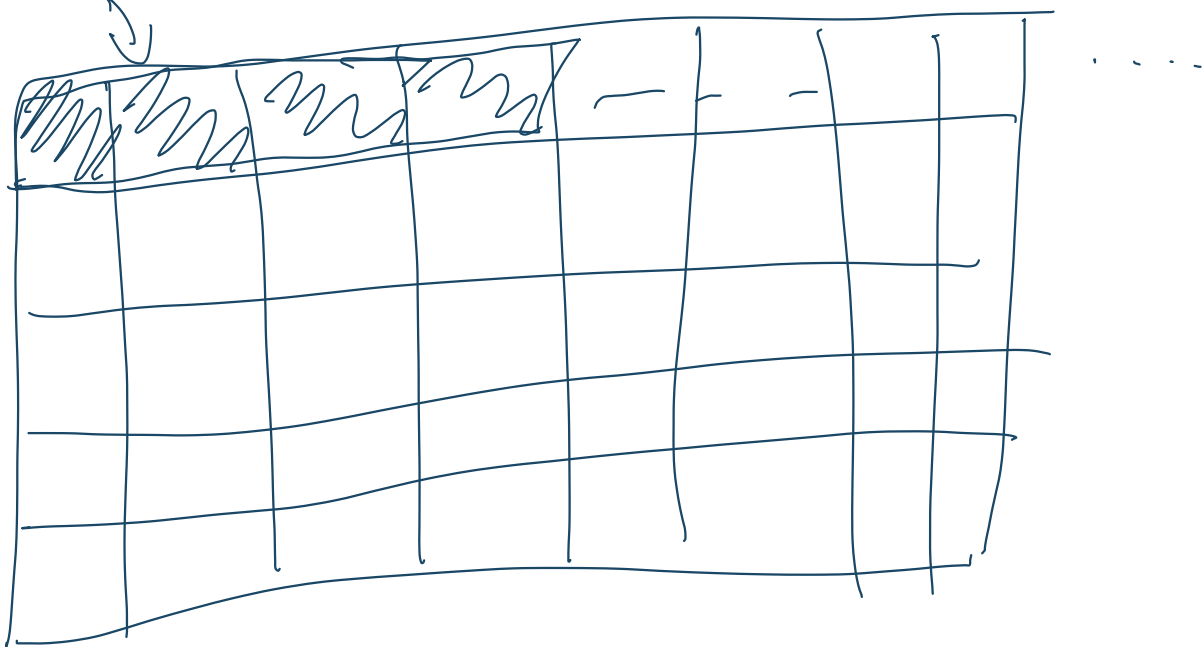
}

$$O(\log(\max(a, b)))$$

$$\text{lcm}(a, b) = \frac{a \cdot b}{\text{gcd}(a, b)}$$

$$\text{lcm}(2^3, 2^5) = \frac{\cancel{2^3} \cdot 2^5}{\cancel{2^3}} = 2^5$$

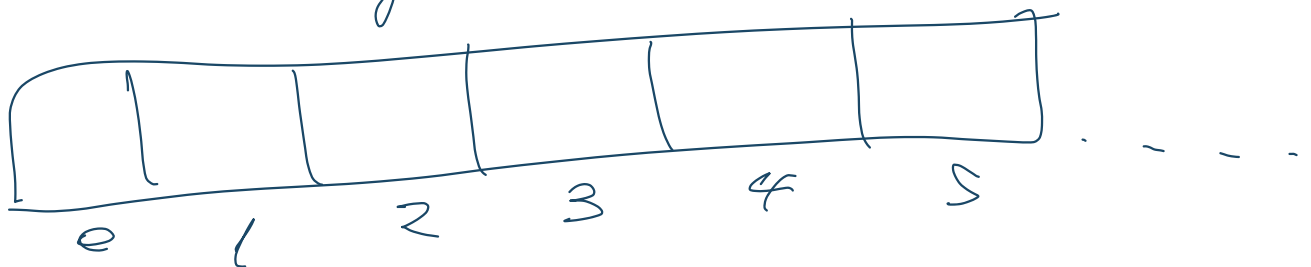
int a; RAM



⋮

□ = 1 byte

1 int = 4 bytes



00

01

02

03

04

⋮

09

10

00

01

02

⋮

09

0A

0B

0C

0D

0E

0F

10

int a;

int *x = &a;

x++;

RAM

x_i

int vector[100];

cout << v[27]; $\rightarrow O(1)$



$*(q + 27)$

$*(q - 1)$

$\downarrow$
 $v[-1]$

